Fiscal Year:	FY 2015	**Task Last Updated:**	FY 12/01/2015
PI Name:	Beaton, Kara Ph.D.		
Project Title:	Forecasting Sensorimotor Adaptability from Baseline Inter-Trial Correlations		
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Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
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Zip Code:	77058-3607	**Congressional District:**	36
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Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bloomberg, Jacob Ph.D. (MENTOR/ NASA Johnson Space Center)		
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Task Description:

POSTDOCTORAL FELLOWSHIP (NOTE ended early due to PostDoc accepting full-time position at Johnson Space Center)
One of the greatest challenges surrounding sensorimotor adaptation to the spaceflight environment is the large variability in symptoms, and corresponding functional impairments, from one crewmember to the next [1-4]. This renders preflight training and inflight countermeasure development difficult, as a one-size-fits-all approach is inappropriate. From an operational perspective, it would be highly advantageous to know ahead of time which crewmembers might have more difficulty adjusting to the novel gravity levels inherent to spaceflight. This type of knowledge could guide individually customized countermeasures to maximize crew safety and mission success. The primary objective of this project was to look for baseline performance metrics that relate to locomotor adaptability. We proposed a novel hypothesis that considers baseline inter-trial correlations, the trial-to-trial fluctuations (noise) in motor performance, as a predictor of individual sensorimotor adaptability. Previous studies have demonstrated that baseline inter-trial correlations are strongly related to oculomotor adaptability [5,6]. The goal of this project was to look for an analogous relationship in the locomotor system.

This work was divided into three specific aims: Aim 1: Predicting locomotor adaptability from inter-trial correlations in baseline step-cadence and heart rate data through retrospective analyses of previously performed treadmill experiments. Aim 2: Predicting locomotor adaptability from baseline inter-trial correlations of step-cadence and heart rate during over-ground (non-treadmill) locomotion. Aim 3: Quantifying the consistency and adaptability of inter-trial correlations.

The initial phases of this project were carried out over a period of three months (March 2015 – June 2015), prior to the PI accepting a full-time position at NASA Johnson Space Center. A subset of this research may continue in the future. During the brief duration of this fellowship, the hardware and software necessary to collect over-ground gait data using wireless motion sensors were set up, the algorithms previously developed for analyzing inter-trial correlations in oculomotor data were adapted for step-cadence data, and baseline treadmill and over-ground step-cadence data were successfully collected and analyzed in several pilot test subjects. Retrospective analyses of inter-trial correlations of baseline step-cadence data from one previously performed treadmill adaptability training experiment were initiated.

Several key data observations have been made to-date. First, inter-trial correlations in baseline step-cadence data from the repeated treadmill and over-ground walking trials were consistent within each test subject. Second, for the previously performed treadmill experiment, the statistical nature of the baseline gait data required that multiple performance metrics be considered to describe inter-trial correlations; hence, new analytical techniques that were not employed in previous oculomotor experiments have been incorporated into this project's overall study design [5,6]. Third, for the previously performed treadmill experiment, baseline inter-trial correlations were unique for each subject.

Future work may include a baseline-only experiment, in which the consistency of baseline inter-trial correlations in step cadence and heart rate will be quantified over several months, and a baseline-plus-adaptation experiment, in which baseline inter-trial correlations in step cadence and heart rate will be compared to locomotor adaptation achieved while walking on a split-belt treadmill.

References:

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2. Davis JR etal (1988) Aviat Space Environ Med 59:1185-1189

3. Locke J (2003) Aviat Space Environ Med 74:101

4. Wood SJ et al (2011) NeuroRehab 29:185-195

5. Wong AL, Shelhamer M (2014) J Neurophys 111:2084-2093

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Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

NOVEMBER 2015 REPORT
Astronauts exhibit significant inter-subject variation in adaptation capabilities upon exposure to altered gravitational levels. Hence the ultimate goal of this project was to uncover individual predictors of sensorimotor adaptability for the purpose of guiding customized countermeasure development to facilitate rapid adaptation to novel sensorimotor challenges. These countermeasures could be in the form of preflight or inflight training prescriptions (e.g., exercises derived from clinical rehabilitation techniques), but would be tailored for each crewmember based on his or her sensorimotor signature revealed by the sensorimotor adaptability predictor. This would enable more efficient use of crew time and provide better outcomes. In the process of seeking these baseline predictors, we hoped to acquire valuable insight as to how motor learning is differentially or similarly accomplished across individuals. Such knowledge has benefits that extend beyond spaceflight research, including, for example, clinical research and rehabilitation techniques for patients with vestibular, cerebellar, and other sensorimotor pathologies. The ability to apply personalized medicine to such individuals would facilitate improved tolerance, enhanced compliance, and faster recoveries. Another significant aspect of this project is the consideration of behavioral performance metrics from two different physiological systems: sensorimotor (locomotor step cadence) and cardiovascular (heart rate). Conventional studies in motor learning typically emphasize the localized effects of a perturbation; their goal is to characterize the aftermath in the context of the response most directly affected. While by no means insignificant, analyzing such responses in isolation limits our understanding of the overall functional capacity of the individual and the collective adaptation process. While impractical to consider every physiological system in a single proposal, adding one non-sensorimotor parameter is the first step in considering these potential predictors more holistically. It is certainly plausible that because of the vast inter-connectedness within the human body, an adaptability predictor discovered in one system may in fact influence, and possibly even stem from, another system. Furthermore, one might posit that an adaptability predictor uncovered in one system might be a more global feature of that individual (e.g., an adaptability phenotype that describes an individual's propensity to retain versus rapidly forget information). For instance, if an individual exhibits strong inter-trial correlations in one system, is it possible that he or she also expresses strong inter-trial correlations in other systems? Such a finding would mean that characterizing baseline performance in one system would not only lead to predictions regarding adaptive capacities in that system, but also predictions about adaptive performance in other systems. By incorporating multiple systems into our study design, we can begin to explore how the CNS (central nervous

	system) might operate at a universal level. Additionally, with the increased trend toward autonomous, portable assessment devices using mobile technology, our tablet-based wireless motion sensors and heart rate monitor provide a convenient platform for beginning such cross-system investigations.
	NOVEMBER 2015 REPORT. During the three months in which this research was funded, the following tasks were completed. First, the tablet-based software and wireless motion sensors that would record all locomotion data (which was previously co-developed in-house by the Principal Investigator) was optimized for this project; this included down-selecting the total number of recorded channels to maximize the high sampling rate necessary for quantifying inter-trial correlations in step cadence data. Additional algorithms were developed to calibrate the raw sensor data, synchronize and interlace the data recorded from the multiple sensors fixed to the left and right legs, and extract the relevant timing information from various phases of the gait cycle. Analysis scripts were compiled to quantify the strength of inter-trial correlations in both the frequency and time domains, including ß (derived from the autocorrelation function), Hurst exponent H (derived from the Hurst rescaled-range analysis), and DFA-a (derived from the de-trended fluctuation analysis). Additional scripts were compiled to generate surrogate baseline datasets to test the statistical likelihood that inter-trial correlations were both present in the original datasets and correlated with adaptive behavior. Baseline walking data (i.e., normal walking at preferred gait speed) was collected both on a treadmill and over-ground in several pilot test subjects to validate these algorithms.
Task Progress:	Second, the study design to complete the three specific aims was refined to include an additional baseline-only experiment that would be completed in Year 1 and a simplification to the baseline-plus-adaptation experiment to be completed in Year 2. The new baseline-only experiment would include the collection of multiple baseline datasets over a period of several months, with the purpose of characterizing consistencies in the baseline performance metrics during repeated testing. The ability to quantify variability over time is significant if any of these metrics are to be considered as potential predictors of individual adaptive capabilities. The baseline-plus-adaptation experiment would now employ a split-belt treadmill, recently acquired in Dr. Bloomberg's laboratory, to carry out the adaptation portion of this experiment. While the locomotor-adaptation platform described in the original research proposal (consisting of a treadmill mounted atop a six degree-of-freedom motion base immersed in a visual surround) provides a challenging multi-sensory stimulus, recent work in the laboratory suggests that the nature of the adaptation itself is too complex for this project. Hence we have revised the experiment for Specific Aim 2 to incorporate adaptation to split-belt treadmill walking. One existing Institutional Review Board (IRB) protocol was modified and accepted, and a new IRB protocol was initiated for these experiments. Finally, retrospective analyses of baseline gait data from other locomotor adaptability training experiments have commenced.
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